Lecture 31

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Contents

1 Polar Representation of Complex Numbers **2**
1.1 Definitions 2
1.2 Multiplication in Polar Form 2
1.3 Rotation by i 2

2 De Moivre's Theorem **2**
2.1 Statement 2
2.2 Proof Sketch 2

3 Example Calculation **3**
3.1 Problem 3
3.2 Solution 3

4 Roots of Complex Numbers **4**
4.1 Finding n-th Roots 4

5 Exponential Notation and Euler's Identity **4**
5.1 Definitions 4
5.2 Equality of Complex Numbers 4
5.3 Euler's Identity 4

6 Roots of Unity **5**
6.1 Cube Roots of Unity ($x^3 = 1$) 5
6.2 General Roots of Unity 5

7 Problems **5**
7.1 Problem 1: Divisibility 5
7.2 Problem 2: Conjugate Equation 5

1 Polar Representation of Complex Numbers

1.1 Definitions

Let $z = r(\cos\theta + i\sin\theta)$. The conjugate is given by:

$$\overline{z} = r(\cos\theta - i\sin\theta) = r(\cos(-\theta) + i\sin(-\theta))$$

1.2 Multiplication in Polar Form

Given two complex numbers:

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$$

$$z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$$

The product is:

$$z_1 \cdot z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$$

1.3 Rotation by i

Since $i = 1(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$, multiplying a number z by i rotates it by $90°$ ($\frac{\pi}{2}$):

$$z \cdot i = r(\cos(\theta + \frac{\pi}{2}) + i\sin(\theta + \frac{\pi}{2}))$$

2 De Moivre's Theorem

2.1 Statement

If $z = r(\cos\theta + i\sin\theta)$ and n is an integer, then:

$$z^n = r^n(\cos(n\theta) + i\sin(n\theta))$$

2.2 Proof Sketch

- For $n \geq 0$, the proof proceeds by mathematical induction.
  - Base cases ($n = 1, n = 2$) satisfy the formula via the multiplication rule.
  - The inductive step assumes the formula for n and proves for $n+1$.

- For negative integers ($n < 0$), let $m = -n$ where $m > 0$.

$$z^{-m} = \frac{1}{z^m} = \frac{\overline{z}^m}{|z|^{2m}}$$

Alternatively, using polar properties directly:

$$z^{-n} = r^{-n}(\cos(-n\theta) + i\sin(-n\theta)) = r^{-n}(\cos(n\theta) - i\sin(n\theta))$$

3 Example Calculation

3.1 Problem

Calculate $(-\sqrt{3} + i)^{11}$.

3.2 Solution

1. **Convert to Polar Form:** Let $z = -\sqrt{3} + i$.

   - Modulus: $r = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = 2$.
   - Angle: Since $x = -\sqrt{3}$ and $y = 1$, the point is in the 2nd quadrant.

   $$z = 2(-\frac{\sqrt{3}}{2} + i\frac{1}{2})$$

   We know $\cos\frac{5\pi}{6} = -\frac{\sqrt{3}}{2}$ and $\sin\frac{5\pi}{6} = \frac{1}{2}$. Thus, $\theta = \frac{5\pi}{6}$.

2. **Apply De Moivre's Theorem:**

$$z^{11} = 2^{11}\left(\cos\left(11 \cdot \frac{5\pi}{6}\right) + i\sin\left(11 \cdot \frac{5\pi}{6}\right)\right)$$

3. **Simplify the Angle:**

$$\frac{55\pi}{6} = \frac{54\pi + \pi}{6} = 9\pi + \frac{\pi}{6} = 8\pi + \pi + \frac{\pi}{6}$$

   This is equivalent to $\pi + \frac{\pi}{6}$ (3rd quadrant).

   - $\cos(\pi + \frac{\pi}{6}) = -\cos\frac{\pi}{6} = -\frac{\sqrt{3}}{2}$
   - $\sin(\pi + \frac{\pi}{6}) = -\sin\frac{\pi}{6} = -\frac{1}{2}$

4. **Final Result:**

$$z^{11} = 2^{11}\left(-\frac{\sqrt{3}}{2} - i\frac{1}{2}\right) = -2^{10}(\sqrt{3} + i)$$

4 Roots of Complex Numbers

4.1 Finding n-th Roots

Given $z = r(\cos\theta + i\sin\theta)$, we want to find w such that $w^n = z$. Let $w = s(\cos\varphi + i\sin\varphi)$. Then $w^n = s^n(\cos n\varphi + i\sin n\varphi)$.

Equating $w^n = z$:

1. $s^n = r \implies s = \sqrt[n]{r}$ (real positive root).
2. $n\varphi = \theta + 2\pi k$ for $k \in \mathbb{Z}$.

The distinct roots are given by $k = 0, 1, \ldots, n-1$:

$$w_k = \sqrt[n]{r}\left(\cos\frac{\theta + 2\pi k}{n} + i\sin\frac{\theta + 2\pi k}{n}\right)$$

5 Exponential Notation and Euler's Identity

5.1 Definitions

- **Exponential Form:** $e^{i\theta} := \cos\theta + i\sin\theta$.
- **Conjugate:** $\overline{z} = re^{-i\theta}$.
- **Multiplication:** $e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$.
- **Powers:** $(re^{i\theta})^n = r^n e^{in\theta}$.

5.2 Equality of Complex Numbers

Let $z = re^{i\theta}$ and $w = se^{i\varphi}$.

$$z = w \iff r = s \wedge \theta - \varphi = 2\pi k \quad (k \in \mathbb{Z})$$

5.3 Euler's Identity

$$e^{i\pi} = \cos\pi + i\sin\pi = -1 + 0$$
$$e^{i\pi} + 1 = 0$$

6 Roots of Unity

6.1 Cube Roots of Unity ($x^3 = 1$)

We solve $x^3 - 1 = 0$:

$$(x-1)(x^2+x+1) = 0$$

- One root is $x_0 = 1$.
- The quadratic equation $x^2 + x + 1 = 0$ yields:

$$x = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm i\sqrt{3}}{2}$$

$$x_1 = -\frac{1}{2} + i\frac{\sqrt{3}}{2} = e^{i\frac{2\pi}{3}}$$

$$x_2 = -\frac{1}{2} - i\frac{\sqrt{3}}{2} = e^{i\frac{4\pi}{3}}$$

6.2 General Roots of Unity

The solutions to $x^n = 1$ are:

$$w_k = \cos\frac{2\pi k}{n} + i\sin\frac{2\pi k}{n} = e^{i\frac{2\pi k}{n}}$$

for $k = 0, 1, \ldots, n-1$.

7 Problems

7.1 Problem 1: Divisibility

Find n such that x^2+x+1 divides $x^{2n}+x^n+1$. **(Note: Setup provided, solution depends on substituting roots of unity).**

7.2 Problem 2: Conjugate Equation

Solve $z^4 = \overline{z}^4$.

Solution: Let $z = re^{i\theta}$. Then $\overline{z} = re^{-i\theta}$.

$$r^4 e^{i4\theta} = r^4 e^{-i4\theta}$$

Assuming $z \neq 0$ (so $r \neq 0$), we divide by r^4:

$$e^{i4\theta} = e^{-i4\theta}$$

This implies the angles differ by a multiple of 2π:

$$4\theta - (-4\theta) = 2\pi k$$

$$8\theta = 2\pi k$$
$$\theta = \frac{2\pi k}{8} = \frac{\pi k}{4}$$

The solutions are complex numbers with arguments equal to multiples of $\frac{\pi}{4}$:

$$z = re^{i\frac{\pi k}{4}}, \quad k \in \{0, 1, \ldots, 7\}, r \in \mathbb{R}$$